

# Linear Transformation HW 1

Section 1.8 # 3, 7, 8, 9, 11, 38

$$3) \left[ \begin{array}{ccc|c} 1 & 0 & -2 & -1 \\ -2 & 1 & 6 & 7 \\ 3 & -2 & -5 & -3 \end{array} \right] \begin{array}{l} 2R_1 + R_2 \rightarrow R_2 \\ -3R_1 + R_3 \rightarrow R_3 \end{array} \left[ \begin{array}{ccc|c} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & -2 & 1 & 0 \end{array} \right]$$

$$2R_2 + R_3 \rightarrow R_3 \left[ \begin{array}{ccc|c} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 5 & 10 \end{array} \right] \quad \frac{1}{5}R_3 \rightarrow R_3$$

$$\left[ \begin{array}{ccc|c} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 2 \end{array} \right] \begin{array}{l} 2R_3 + R_1 \rightarrow R_1 \\ -2R_3 + R_2 \rightarrow R_2 \end{array} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Thus, the vector  $\vec{x} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$  is the preimage of

$\vec{b}$  under the Transformation  $T(\vec{x}) = A\vec{x}$ .

$$7) A_{6 \times 5}. T: \mathbb{R}^5 \rightarrow \mathbb{R}^6, T(\vec{x}) = A\vec{x}$$

The number of columns determines the domain and the number of rows determines the codomain. Since  $A$  is  $6 \times 5$ ,  $a = 5$  and  $b = 6$ .

$$8) T: \mathbb{R}^4 \rightarrow \mathbb{R}^5. A \text{ needs to be } 5 \times 4.$$

Remember columns  $\rightarrow$  domain  
rows  $\rightarrow$  codomain

$$9) \left[ \begin{array}{cccc|c} 1 & -4 & 7 & -5 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 2 & -6 & 6 & -4 & 0 \end{array} \right] \quad -2R_1 + R_3 \rightarrow R_3$$

$$\left[ \begin{array}{cccc|c} 1 & -4 & 7 & -5 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 2 & 8 & 6 & 0 \end{array} \right] \begin{array}{l} 4R_2 + R_1 \rightarrow R_1 \\ -2R_2 + R_3 \rightarrow R_3 \end{array} \left[ \begin{array}{cccc|c} 1 & 0 & -9 & 7 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} x_1 - 9x_3 + 7x_4 = 0 \\ x_2 - 4x_3 + 3x_4 = 0 \end{array}$$

$$\begin{array}{l} \rightarrow x_1 = 9x_3 - 7x_4 \\ x_2 = 4x_3 - 3x_4 \\ x_3 = t \text{ (free)} \\ x_4 = s \text{ (free)} \end{array}$$

$$\vec{x} = \begin{bmatrix} 9t - 7s \\ 4t - 3s \\ t \\ s \end{bmatrix}$$

$$11) \left[ \begin{array}{cccc|c} 1 & -4 & 7 & -5 & -1 \\ 0 & 1 & -4 & 3 & 1 \\ 2 & -6 & 6 & -4 & 0 \end{array} \right] \quad -2R_1 + R_3 \rightarrow R_3$$

$$\left[ \begin{array}{cccc|c} 1 & -4 & 7 & -5 & -1 \\ 0 & 1 & -4 & 3 & 1 \\ 0 & 2 & 6 & 4 & 2 \end{array} \right] \quad \begin{array}{l} 4R_2 + R_1 \rightarrow R_1 \\ -2R_2 + R_3 \rightarrow R_3 \end{array}$$

$$\left[ \begin{array}{cccc|c} 1 & 0 & -9 & 7 & 3 \\ 0 & 1 & -4 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \leftarrow \text{since this is 0, yes it's in the range! In fact, infinitely vectors get mapped to it. They have the form:}$$

$$x_1 = 9x_3 - 7x_4 + 3$$

$$x_2 = 4x_3 - 3x_4 + 1$$

$$x_3 = \text{free}$$

$$x_4 = \text{free}$$

$$\vec{x} = \begin{bmatrix} 9s - 7t + 3 \\ 4s - 3t + 1 \\ s \\ t \end{bmatrix} \quad s, t \in \mathbb{R}$$

\* I'm just doing the same EROs in the same order, as I did in #9. The only thing that changes is the augmented column. :)

$$38) T(\vec{x}) = A\vec{x} + \vec{b} \quad (\text{affine transformation})$$

Since it's Not linear, it will violate one (or both) of these 2 properties:

$$T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$$

$$T(c\vec{x}) = cT(\vec{x})$$

$$\begin{aligned} \text{Not } (\Rightarrow) \quad & \left\{ \begin{array}{l} T(\vec{x} + \vec{y}) = A(\vec{x} + \vec{y}) + \vec{b} = A\vec{x} + A\vec{y} + \vec{b} \\ T(\vec{x}) + T(\vec{y}) = A\vec{x} + \vec{b} + A\vec{y} + \vec{b} = A\vec{x} + A\vec{y} + 2\vec{b} \end{array} \right. \end{aligned}$$

↑ Not linear since the only way for  $\vec{b} = 2\vec{b}$  is if  $\vec{b} = \vec{0}$ . Thus, it's only linear when  $\vec{b} = \vec{0}$ .

PDF Problems → proof for 333 kids (lazy way out)

1) Proof: Let  $T$  be the transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by  $T(\vec{x}) = A\vec{x}$  where  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ .

Consider  $T(\vec{x} + \vec{y})$ , where  $\vec{x}, \vec{y} \in \mathbb{R}^2$ . Then,

$$T(\vec{x} + \vec{y}) = A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} \quad \text{by theorem 5 (section 1.4). Also,}$$

$$A\vec{x} + A\vec{y} = T(\vec{x}) + T(\vec{y}).$$

Now let  $c \in \mathbb{R}$  be a scalar and consider  $T(c\vec{x})$ ,  $\vec{x} \in \mathbb{R}^2$ . Then,

$T(c\vec{x}) = A(c\vec{x}) = cA\vec{x}$  by Theorem 5 from section 1.4. Also,  $CA(\vec{x}) = cT(\vec{x})$ . Thus since  $T$  satisfies the 2 properties  $T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$  and  $T(c\vec{x}) = cT(\vec{x})$ , then  $T$  is a linear transformation.  $\square$

$$\begin{aligned} \text{ol: } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} &= \begin{bmatrix} a(x_1 + y_1) + b(x_2 + y_2) \\ c(x_1 + y_1) + d(x_2 + y_2) \end{bmatrix} \\ &= \begin{bmatrix} ax_1 + ay_1 + bx_2 + by_2 \\ cx_1 + cy_1 + dx_2 + dy_2 \end{bmatrix} \\ &= \begin{bmatrix} (ax_1 + bx_2) + (ay_1 + by_2) \\ (cx_1 + dx_2) + (cy_1 + dy_2) \end{bmatrix} \\ &= \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix} + \begin{bmatrix} ay_1 + by_2 \\ cy_1 + dy_2 \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \end{aligned}$$

So,  $A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y}$

Now,  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} sx_1 \\ sx_2 \end{bmatrix} = \begin{bmatrix} asx_1 + bsx_2 \\ csx_1 + dsx_2 \end{bmatrix} = \begin{bmatrix} s(ax_1 + bx_2) \\ s(cx_1 + dx_2) \end{bmatrix}$

$c$  was already used, so I used an  $s$  for my scalar.

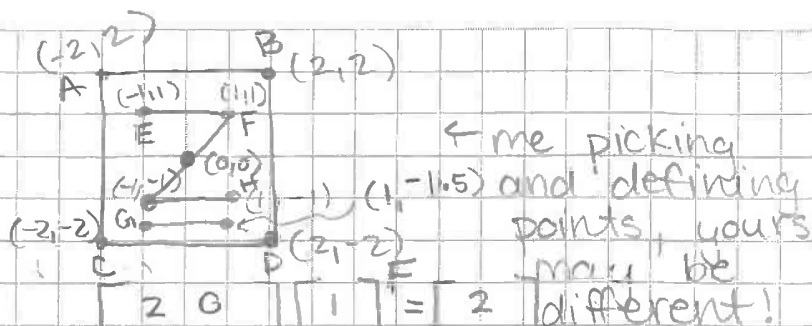
$$\begin{aligned} &= s \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix} \\ &= s \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

So,  $A(c\vec{x}) = cA\vec{x}$ .  $\square$

2) It is Not a linear transformation as  $\vec{0}$  is Not mapped to  $\vec{0}$ . Why does  $\vec{0}$  need to be mapped to  $\vec{0}$ ? Look @ the 2nd property:

$A(c\vec{x}) = cA\vec{x}$ . This is true for all scalars  $c$ !  
Thus when  $c = 0$ , we have  $A(\vec{0}) = A(\vec{0})$   
 $= 0 \cdot A\vec{x}$   
 $= \vec{0}$ .

$$3) C = \begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix}$$



a)

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1.5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1.5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} -2 \\ -2 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$

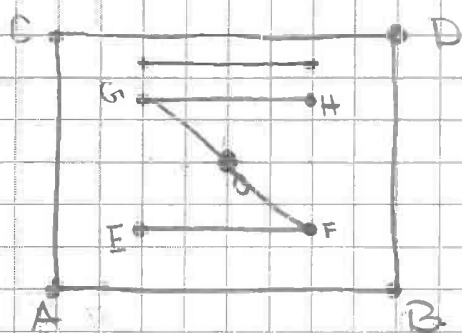
$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1.5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1.5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} 1 \\ -1.5 \end{bmatrix} = \begin{bmatrix} 2 \\ 2.25 \end{bmatrix} \quad \begin{bmatrix} 2 & 0 \\ 0 & -1.5 \end{bmatrix} \begin{bmatrix} -1 \\ -1.5 \end{bmatrix} = \begin{bmatrix} -2 \\ 2.25 \end{bmatrix}$$

OK! Now I can sketch w/ confidence!



← Image under C.

b) The image got stretched in the x-direction by a factor of 2 (i.e. the image got 2 times wider). It also got 1.5 times taller but it was also reflected across the x-axis (i.e. it got flipped upside down).

c) I labeled my points and simply calculated  $C\vec{x}$  to see what happened to each point under C. I knew I was right b/c I'm amazing! muwahaha.

$$4) e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

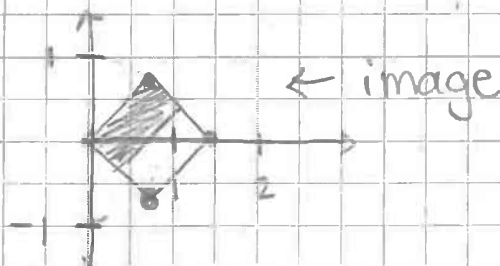


a) where does  $e_1$  go when rotated by  $-\pi/4$ ? It will be sent to  $\begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}$  (unit circle)

$e_2$  gets sent to  $\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$ . Thus the standard

matrix  $A$  will be:  $A = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

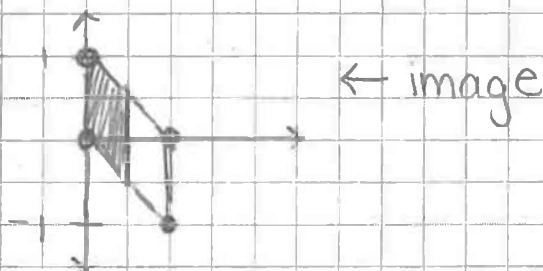
$$\langle 1, 1 \rangle \mapsto \langle \sqrt{2}/2 + \sqrt{2}/2, -\sqrt{2}/2 + \sqrt{2}/2 \rangle = \langle \sqrt{2}, 0 \rangle$$



b)  $e_1 \mapsto \langle 1-0, 0-1 \rangle = \langle 1, -1 \rangle$ ,  $e_2 \mapsto e_2$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

$$\langle 1, 1 \rangle \mapsto \langle 1+0, -1+1 \rangle = \langle 1, 0 \rangle$$



c)  $e_1 \mapsto \langle -1, 0 \rangle \mapsto \langle 0, -1 \rangle$

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

reflect

rotate

$e_2 \mapsto \langle 0, 1 \rangle \mapsto \langle -1, 0 \rangle$

$$\langle 1, 1 \rangle \mapsto \langle -1, 1 \rangle \mapsto \langle -1, -1 \rangle$$

